

Twist averaging is not enough: virtual-path windings and the exact zero-transport projector for the quantum spin ice gauge sector

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Abstract

This note is the theory backbone of the accompanying program: turning a small periodic quantum spin ice (QSI) cluster into a quantitative laboratory for the emergent U(1) gauge sector by removing the torus-winding processes that have no infinite-lattice analogue. An earlier version of this document proved a *Winding-Projection Theorem* asserting that averaging over twisted boundary conditions removes those processes exactly, for every closed-system observable. **That theorem is false, and we retract it here.** Its error is a single, instructive one: it assigned each emergent ring exchange the holonomy of its *loop*, whereas the twist phase of a perturbative process is the product of the phases of the bond terms the process *actually uses* – a property of virtual *paths*, not loops. We prove the correct structural result, the *Matched-Pair Factorization* (theorem 1): a ring exchange over a cycle $\mathcal{C}$ is generated by the several perfect matchings of $\mathcal{C}$ into dimer moves, and its twisted coupling is the sum of $e^{-i\mathbf{N}\cdot\boldsymbol{\varphi}}$ over matchings, with $\sum \mathbf{N} = \mathbf{w}_{\mathcal{C}}$ the loop winding. Two corollaries follow and are the honest verdict on twist averaging: (A) every torus-winding ring exchange has a matching with zero net wrap, so *any* twist average of the Hamiltonian – continuum, grid, or corners – retains a nonzero fraction (one half for a two-matching loop) of every spurious coupling; and (B) averaging *observables* is weaker still, cancelling only the parts odd in the spurious coupling. We then construct the object that does work: the *zero-transport projector* (theorem 2), which keeps only processes whose net charge transport $\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N} \cdot \mathbf{L}$ vanishes – the gauge-invariant label to which a uniform emergent vector potential couples. It annihilates every winding ring exchange at all orders while preserving every contractible one, and, being a statement about operators rather than spectra, cleans not only H but every ice-manifold response operator. A tight-binding toy solvable by hand exhibits all three statements. We close with the specialization to QSI and the role of the projector in the π -flux-beyond-QMC campaign.

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1 Plain-language summary

We simulate a magnet by cutting a small block out of the crystal and gluing opposite faces together, so a spin leaving one face re-enters the opposite one. The glued block is a three-dimensional torus, and diagonalizing the spin Hamiltonian on it is *exact diagonalization on a periodic cluster*.

The gluing creates fake shortcuts: closed loops of spin flips that exist only because the faces are identified, with no counterpart on the infinite crystal. In QSI these fake loops are especially damaging because they are *short* – they enter the low-energy (gauge) dynamics at a lower order in the coupling than the genuine ring exchange, so they can dominate exactly the quantity one wants.

The natural idea is to thread an Aharonov–Bohm flux φ through each wrap-around direction, a “twist,” and average the results over φ . A loop that wraps picks up a phase; average over the dial and the phase seems to average to zero. **This intuition is wrong**, and the reason is the point of this note. An emergent ring exchange is not a single connected walk around the loop; it is a product of local moves, and the same net loop can be produced by several distinct arrangements of those moves. One of those arrangements always uses only interior (non-wrapping) moves – so it carries *no* twist phase, survives any average, and leaves the fake loop half-alive. Worse, in practice one averages a specific number like the specific heat, which is a nonlinear function of the energies; averaging it cancels even less.

The cure is to look not at which loop a process draws, but at how much electric charge it *transports* across the cluster. A genuine local ring exchange transports nothing; a fake wrapping loop necessarily transports charge from one side to the other. Keeping only the zero-transport processes deletes every fake loop exactly, at every order, while leaving every genuine one untouched – and it does so for every physical quantity at once, because it is a rule about the processes themselves, not about any one observable. The rest of this note makes each italicized claim a theorem.

2 Setup: the cluster torus and the ice manifold

The XXZ pyrochlore Hamiltonian in the local frames is

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad |J_{\pm}| \ll J_{zz}. \quad (1)$$

The cluster is a set of N sites with periodic identifications generated by three translation vectors collected as the rows of a matrix $\mathbf{L} \in \mathbb{Z}^{3 \times 3}$; the position space is the flat torus $T^3 = \mathbb{R}^3 / \mathbf{L}\mathbb{Z}^3$, with first homology $H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3$. For a nearest-neighbour bond (i, j) let $\mathbf{n}_{ij} = \text{round}[(\mathbf{r}_j - \mathbf{r}_i)\mathbf{L}^{-1}] \in \mathbb{Z}^3$ be its *wrap vector*: $\mathbf{n}_{ij} = \mathbf{0}$ for an interior bond, nonzero for a boundary-crossing one.

For $|J_{\pm}| \ll J_{zz}$ the low-energy manifold is the space of 2-in-2-out (ice-rule) configurations, $\dim = N_{\text{ice}}$; degenerate perturbation theory in the transverse term produces an effective ring Hamiltonian H_{eff} acting on it. On the infinite lattice the shortest ice-preserving loop is the six-site hexagon, giving the standard photon coupling $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$. On the torus the identifications also admit shorter ice-preserving cycles that wind; the dominant ones are four-site loops, second order in $J_{\pm}$, coupling $g_4 = 4J_{\pm}^2/J_{zz} \gg g_{\text{hex}}$.

3 A twist is a flat $U(1)$ connection; its holonomy

A twist $\varphi \in [0, 2\pi)^3$ is inserted by attaching a phase to each transverse hop according to the bond it uses,

$$H_{\perp}(\varphi) = -J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- e^{-i\mathbf{n}_{ij} \cdot \varphi} + S_i^- S_j^+ e^{+i\mathbf{n}_{ij} \cdot \varphi}), \quad (2)$$

in the *minimal (boundary-bond) gauge*, where only wrapping bonds carry phases. The Ising term is untouched, S_{tot}^z is conserved at every φ , and $\varphi = \mathbf{0}$ recovers eq. (1). Because the connection is flat, its only gauge-invariant content is holonomy: for a closed sequence of hops with total wrap $\mathbf{w} = \sum_a \mathbf{n}_{ia ia+1} \in \mathbb{Z}^3$ the accumulated phase is the Wilson loop $e^{-i\mathbf{w} \cdot \varphi}$, a function of the homology class alone.

The tempting – and false – inference is that this makes each emergent ring exchange carry $e^{-i\mathbf{w} \cdot \varphi}$. The next section shows why it does not: a ring exchange is not a single closed hop sequence.

4 The engine: operators are sums over virtual paths

Every quantity computed from $H_{\perp}(\varphi)$ – matrix elements of H_{eff} , the resolvent $(z - H_{\perp})^{-1}$, the heat kernel $e^{-\beta H_{\perp}}$ – is, by expanding in $H_{\perp}$, a sum over sequences of elementary hops, each sequence weighted by the product of its hop amplitudes. Under (2) each hop ($i \rightarrow j$) contributes a phase $e^{-i\mathbf{n}_{ij} \cdot \varphi}$. Hence a process that starts and ends in the same configuration and uses hops of total wrap $\mathbf{N}$ carries the phase $e^{-i\mathbf{N} \cdot \varphi}$.

The crucial point is that an *operator* on the ice manifold – an emergent ring exchange – is not one such sequence. The ring-exchange operator over an ice-preserving cycle $\mathcal{C} = (abcd \dots)$,

$$O_{\mathcal{C}} = (S_a^+ S_b^- S_c^+ S_d^- \dots), \quad (3)$$

is a product of commuting local dimer moves, and it is generated in perturbation theory by *every* factorization of $\mathcal{C}$ into disjoint nearest-neighbour dimer moves – the *perfect matchings* of the cycle. Different matchings use different bonds, hence different total wraps $\mathbf{N}$. This is the entire content of the correction to section 3.

5 The Matched-Pair Factorization Theorem

Theorem 1 (Matched-Pair Factorization). *Let $\mathcal{C}$ be an ice-preserving simple cycle of even length $2m$ on the cluster, with winding $\mathbf{w}_{\mathcal{C}}$. In the minimal gauge (2), at the leading perturbative order at which $O_{\mathcal{C}}$ appears, its coefficient in $H_{\text{eff}}(\varphi)$ is*

$$c_{\mathcal{C}}(\varphi) = -\frac{g_{\mathcal{C}}}{M} \sum_{\mu=1}^M e^{-i\mathbf{N}_{\mu} \cdot \varphi}, \quad \mathbf{N}_{\mu} = \sum_{(ij) \in \text{matching } \mu} \mathbf{n}_{ij}, \quad (4)$$

where the sum runs over the M perfect matchings of $\mathcal{C}$ into disjoint dimer moves, $g_{\mathcal{C}} > 0$ is the (real, gauge-fixed) bare coupling, and the matching wraps obey $\mathbf{N}_1 + \mathbf{N}_2 = \mathbf{w}_{\mathcal{C}}$ for a two-matching loop. Only the individual matching wraps $\mathbf{N}_{\mu}$ enter; the loop winding constrains only their sum. In particular, the gauge-invariant magnitude for a two-matching loop ($M = 2$) is

$$|c_{\mathcal{C}}(\varphi)| = \frac{g_{\mathcal{C}}}{2} |1 + e^{-i(\mathbf{N}_2 - \mathbf{N}_1) \cdot \varphi}| = g_{\mathcal{C}} \left| \cos \frac{\mathbf{w}_{\mathcal{C}} \cdot \varphi}{2} \right|, \quad (5)$$

using $\mathbf{N}_2 - \mathbf{N}_1 = \pm \mathbf{w}_{\mathcal{C}}$.

Proof. At the leading order, $O_{\mathcal{C}}$ is produced only by processes that flip each of the $2m$ spins of $\mathcal{C}$ exactly once, with no other flips; these are exactly the products of m disjoint dimer moves covering $\mathcal{C}$, i.e. its perfect matchings. A cycle of even length has $M \geq 2$ matchings (for a 4-cycle, $M = 2$: $\{ab, cd\}$ and $\{bc, da\}$). Each matching μ is a product of m commuting hops; in the minimal gauge its phase is the product $e^{-i\mathbf{N}_{\mu} \cdot \varphi}$ with $\mathbf{N}_{\mu}$ the sum of the constituent bond wraps. The real amplitudes of all matchings coincide (equal denominators and combinatorial weight at leading order), giving the prefactor $g_{\mathcal{C}}/M$. Finally, $\mathbf{N}_1 + \mathbf{N}_2$ for the two matchings of a 4-cycle traverses each bond of $\mathcal{C}$ once, i.e. equals the loop wrap $\mathbf{w}_{\mathcal{C}}$; hence $\mathbf{N}_2 - \mathbf{N}_1 \equiv \mathbf{w}_{\mathcal{C}} \pmod{2}$ and (5) follows. $\square$

Remark 1 (Where the old proof failed). The retracted Winding-Projection Theorem assumed $M = 1$: that $O_{\mathcal{C}}$ carries the single phase $e^{-i\mathbf{w}_{\mathcal{C}} \cdot \varphi}$. Equation (4) shows the operator lives in *several* twist-Fourier sectors $\mathbf{N}_{\mu}$ at once; the holonomy $e^{-i\mathbf{w} \cdot \varphi}$ appears only as the *relative* phase of the matchings, (5). Everything below is a consequence.

6 Corollary A: twist-averaging the Hamiltonian keeps half

Corollary 1 (Half-survival). *Suppose $\mathcal{C}$ is a torus-winding ice loop possessing a matching μ_0 with $\mathbf{N}_{\mu_0} = \mathbf{0}$ (all its dimer moves interior). Then for any averaging measure $d\nu(\varphi)$ on the twist torus – the continuum $\int d^3\varphi/(2\pi)^3$, any regular grid, or the eight corners $\{0, \pi\}^3$ –*

$$\overline{c_{\mathcal{C}}} = \int c_{\mathcal{C}}(\varphi) d\nu(\varphi) = -\frac{g_{\mathcal{C}}}{M} \left(1 + \sum_{\mu \neq \mu_0} \widehat{\nu}(\mathbf{N}_{\mu}) \right), \quad (6)$$

where $\widehat{\nu}(\mathbf{N}) = \int e^{-i\mathbf{N} \cdot \varphi} d\nu$. The μ_0 term $-g_{\mathcal{C}}/M$ survives untouched, and for a two-matching loop this is exactly one half of the bare coupling. Twist averaging of H_{eff} therefore cannot annihilate $O_{\mathcal{C}}$; at best it removes the $M - 1$ wrapping matchings.

Proof. $\widehat{\nu}(\mathbf{0}) = 1$ for any normalized measure, so the $\mathbf{N}_{\mu_0} = \mathbf{0}$ term is invariant under averaging. The other terms are multiplied by $\widehat{\nu}(\mathbf{N}_{\mu})$, of modulus ≤ 1 . $\square$

Proposition 1 (The hypothesis always holds on the production clusters). *On the $1 \times 1 \times 1$ cubic (16-site) and $2 \times 2 \times 2$ FCC (32-site) pyrochlore clusters, every ice-preserving winding four-loop and every wrapping hexagon has a matching with $\mathbf{N}_{\mu_0} = \mathbf{0}$, whereas every contractible hexagon has all matchings with $\mathbf{N}_{\mu} = \mathbf{0}$.*

This is verified by exhaustive enumeration (path-resolved Schrieffer–Wolff, tracking $\mathbf{N}$ per virtual path): on the cubic cluster all 36 four-loops and all 48 wrapping hexagons retain $\frac{1}{2}$ their coupling under the eight-corner operator average, while all 16 contractible hexagons are preserved in full; the FCC cluster gives the same with 24, 96, 32. Thus twist averaging leaves the artifact at the $g_4/2$ scale – a fact confirmed spectrally: the corner-averaged 16-site low- T peak stalls at $0.018 J_{zz} \approx g_4/2$, never reaching $g_{\text{hex}} = 0.012 J_{zz}$.

Corollary 2 (Corner switch-off, not sign-flip). *At a corner $\varphi = \pi \mathbf{p}$, $\mathbf{p} \in \{0, 1\}^3$, (5) gives $|c_c| = g_c$ if $\mathbf{w}_c \cdot \mathbf{p}$ is even and 0 if odd: a winding loop's coupling is switched fully off at half the corners, not sign-reversed. The eight-corner average is therefore a loop-deactivation histogram, not a cancellation.*

7 Corollary B: averaging observables is weaker still

In practice one does not average H_{eff} ; one averages an *observable* – the specific heat $C(T, \varphi)$, the free energy, a structure factor – and these are nonlinear in H_{eff} .

Corollary 3 (Even powers survive). *Write the spurious coupling of a single winding loop as $g(\varphi) = g_c \cos(\mathbf{w} \cdot \varphi/2)$ [eq. (5)] and let $\mathcal{O}(g)$ be any closed-system observable, analytic in g near the physical value. Its eight-corner (or continuum) average retains every even power of g :*

$$\overline{\mathcal{O}} = \sum_{k \geq 0} \mathcal{O}_{2k} \overline{g^{2k}}, \quad \overline{g^{2k}} = g_c^{2k} \overline{\cos^{2k}} \neq 0, \quad (7)$$

because $g(\varphi)$ is even about each corner. Odd powers, and only they, can cancel. Since a winding loop enters observables already at $O(g^2)$ – e.g. a gap $2\sqrt{\varepsilon^2 + g^2}$ about a genuine scale $\varepsilon \sim g_{\text{hex}}$ – and $g_c = g_4 \gg g_{\text{hex}}$, the surviving even-power contamination dominates the physical scale.

Thus the observable average is strictly weaker than the operator average of section 6: on 16 sites the low- T peak moves only from $0.037 J_{zz}$ (bare) to $0.031 J_{zz}$ (corner-averaged observable), against $0.018 J_{zz}$ for the corner-averaged operator – neither reaching g_{hex} .

8 The zero-transport projector

The correct invariant is not homology but transport. Define, for a process taking ice configuration c to c' , the *net charge transport*

$$\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N} \cdot \mathbf{L}, \quad \boldsymbol{\rho} = \sum_{i \text{ raised}} \mathbf{r}_i - \sum_{j \text{ lowered}} \mathbf{r}_j, \quad (8)$$

where $\boldsymbol{\rho}$ is the dipole of the flipped-site set (the operator's intrinsic displacement) and $\mathbf{N} \cdot \mathbf{L}$ the contribution of any wraps. $\boldsymbol{\delta}$ is the physical displacement of the transported S^z (boson) density and is gauge invariant: it is exactly what a *uniform* emergent vector potential couples to. In the smooth gauge $\theta_{ij} = \mathbf{A} \cdot \mathbf{d}_{ij}$ with $\mathbf{L}\mathbf{A} = \boldsymbol{\varphi}$, every process carries $e^{-i\mathbf{A} \cdot \boldsymbol{\delta}}$, and averaging over the extended twist period is projection onto $\boldsymbol{\delta} = \mathbf{0}$.

Theorem 2 (Zero-transport projector). *The projector*

$$H_{\text{eff}}^{\boldsymbol{\delta}} = \sum_{\text{processes: } \boldsymbol{\delta}=\mathbf{0}} (\text{that process}) \quad (9)$$

annihilates every torus-winding ice ring exchange, at every order in $J_{\pm}$, and preserves every contractible one, provided the operator dipole is too short to bridge a cluster period ($|\boldsymbol{\rho}| < \min_{\mathbf{w} \neq \mathbf{0}} |\mathbf{w} \cdot \mathbf{L}|$), which holds for the four-loops and hexagons of the production clusters.

Proof. Winding loops die. A matching of a ring loop transports the flipped bosons by at most its diameter, $|\boldsymbol{\rho}| \leq (\text{loop extent})$; for a 4-cycle this is $\leq 2d_{\text{NN}} = a/\sqrt{2}$, for a hexagon $\leq 3d_{\text{NN}}$. A

winding loop has $\mathbf{w} \neq \mathbf{0}$, so its two matchings' transports differ by $\mathbf{w} \cdot \mathbf{L}$, with $|\mathbf{w} \cdot \mathbf{L}| \geq a$ (cubic) or $\sqrt{2}a$ (FCC): since $|\boldsymbol{\rho}| < |\mathbf{w} \cdot \mathbf{L}|$, no matching can reach $\boldsymbol{\delta} = \mathbf{0}$. Higher-order dressings shift $\boldsymbol{\rho}$ only by full lattice vectors, which the short operator dipole cannot cancel; hence $\boldsymbol{\delta} \neq \mathbf{0}$ for a winding loop at every order.

Contractible loops survive. A contractible ice loop has $\mathbf{w} = \mathbf{0}$ and, being a genuine bulk plaquette (a regular [111] hexagon), has vanishing alternating displacement sum, $\boldsymbol{\rho} = \mathbf{0}$, for each matching. Thus $\boldsymbol{\delta} = \mathbf{0}$ and the process is kept in full. $\square$

Remark 2 (Operator, not spectrum). theorem 2 is a rule on processes, so it cleans *every* ice-manifold operator simultaneously – H_{eff} and every induced response operator alike. This is what upgrades the projector from a spectral fix to a definition of the bulk gauge sector of a torus, and is the basis of the response program of the main paper.

Remark 3 (One honest bias). The projector also deletes any *contractible* process carrying a nonzero transport dipole ($\boldsymbol{\rho} \neq \mathbf{0}$), possible for non-centrosymmetric loops at order $J_{\pm}^4$ and beyond. Within the nearest-neighbour XXZ model these are $O(J_{\pm}^4/J_{zz}^3)$, parametrically below g_{hex} ; we flag rather than hide them.

9 A toy model solvable by hand

Take the two-dimensional space spanned by a single ice loop's two ice configurations $\{|+\rangle, |-\rangle\}$, connected by a winding 4-loop with two matchings, and coupled to a genuine (contractible) process of strength ε . By theorem 1 the winding block at twist φ (one relevant component) is

$$H_{\text{eff}}(\varphi) = \begin{pmatrix} 0 & \varepsilon + \frac{g_4}{2}(1 + e^{-i\varphi}) \\ \text{h.c.} & 0 \end{pmatrix}, \quad (10)$$

with $\mathbf{N}_1 = 0$ (interior matching), $\mathbf{N}_2 = 1$ (wrapping matching), so the winding coupling is $g(\varphi) = \frac{g_4}{2}(1 + e^{-i\varphi})$.

Operator average (corollary 1): $\bar{g} = \frac{g_4}{2}(1 + \overline{e^{-i\varphi}}) = \frac{g_4}{2}$ for any average with $\overline{e^{-i\varphi}} = 0$ (continuum, or the two “corners” $\varphi \in \{0, \pi\}$). Half survives.

Observable average (corollary 3): the gap is $\Delta(\varphi) = 2|\varepsilon + g(\varphi)|$, and $\bar{\Delta}$ over $\varphi \in \{0, \pi\}$ is $|\varepsilon + g_4| + |\varepsilon|$ – the g_4 term never cancels. Even less is removed than by the operator average.

Zero-transport projection (theorem 2): both winding matchings carry $\boldsymbol{\delta} \neq \mathbf{0}$ (the interior matching a bare operator dipole $\boldsymbol{\rho} \neq \mathbf{0}$, the wrapping matching $\boldsymbol{\rho} + \mathbf{L} \neq \mathbf{0}$), while the genuine contractible process has $\boldsymbol{\delta} = \mathbf{0}$. Keeping $\boldsymbol{\delta} = \mathbf{0}$ retains ε and deletes *both* winding matchings, giving the clean $2|\varepsilon|$. Only this scheme reaches the physical answer.

10 Specialization to QSI, and role in the campaign

For the XXZ pyrochlore the winding loops are the second-order four-site cycles ($g_4 = 4J_{\pm}^2/J_{zz}$) and the wrapping hexagons; the contractible loops are the genuine [111] hexagons carrying the photon ($g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$). theorem 2 removes the former at all orders and keeps the latter at bulk density, so the projected effective Hamiltonian on the 2970-state ice manifold of the $2 \times 2 \times 2$ FCC cluster is a clean, fully diagonalizable model of the emergent gauge sector.

Because the couplings scale as $J_{\pm}^k$ exactly, one effective-Hamiltonian build serves every coupling and both signs of $J_{\pm}$. This is the enabling step of the accompanying program: benchmark the projected 0-flux ($J_{\pm} > 0$) gauge thermodynamics against sign-problem-free quantum Monte Carlo,

then cross into the π -flux ($J_{\pm} < 0$) sector – the QMC dead zone of the cerium pyrochlores – where the transport-projected response operators yield the flux-locked thermodynamic discriminant of the emergent photon.

11 Verdict

Twist averaging, as a means of removing torus-winding contamination from QSI exact diagonalization, *does not work*: averaging the Hamiltonian keeps half of every winding coupling (corollary 1), and averaging observables keeps even more (corollary 3). Both failures trace to a single misidentification – assigning a loop’s holonomy to an operator that is in fact a sum over virtual paths (theorem 1). The object that succeeds is the zero-transport projector (theorem 2): gauge invariant, exact at all orders for the winding loops, transparent about its one higher-order bias, and – because it acts on processes rather than spectra – a cleaning of the entire gauge sector, Hamiltonian and responses alike. That is what makes a 32-site pyrochlore a quantitative laboratory for π -flux quantum spin ice.